

ExamNoteX

NUCLEI

Nucleus

Definition:

- The central positively charged part of an atom is called the nucleus.
- It contains protons and neutrons.

Nuclear Force

Definition:

The strong attractive force acting between nucleons inside nucleus is called nuclear force.

Characteristics:

- Very strong force
- Short range force
- Independent of charge
- Attractive in nature

Atomic Number

Atomic number is the number of protons present in the nucleus.

Formula:

Z = Number of protons

Mass Number

A = Number of protons + Number of neutrons

Mass Defect

Definition:

The difference between the total mass of nucleons and actual mass of nucleus is called mass defect.

Formula:

$$\Delta m = [Zm_p + (A - Z)m_n] - M$$

Where:

- Δm = Mass defect
- m_p = Mass of proton
- m_n = Mass of neutron
- M = Actual mass of nucleus

Binding Energy

Definition:

The energy required to separate nucleons completely from nucleus is called binding energy.

Formula:

$$E = \Delta mc^2$$

Important Point:

The greater the binding energy, the greater is the stability of the nucleus.

Binding Energy per Nucleon

Binding Energy per nucleon = Total binding energy / Mass number

Radioactivity

Definition:

The spontaneous disintegration of an unstable nucleus accompanied by emission of radiation is called radioactivity.

Types of Radiations

Alpha Rays:

Alpha rays consist of helium nuclei and carry positive charge.

Beta Rays:

Beta rays consist of fast moving electrons.

Gamma Rays:

Gamma rays are electromagnetic radiations with high penetrating power.

Half Life

Definition:

Half life is the time required for half of the radioactive nuclei to decay.

Formula:

$$T_{1/2} = 0.693 / \lambda$$

Where:

- $T_{1/2}$ = Half life
- λ = Decay constant

Nuclear Fission

Definition:

The splitting of a heavy nucleus into lighter nuclei with release of large amounts of energy is called nuclear fission.

Example:

Uranium-235 undergoes nuclear fission.

Nuclear Fusion

Definition:

The combination of lighter nuclei to form a heavier nucleus with release of energy is called nuclear fusion.

Example:

Energy produced in the Sun is due to nuclear fusion.

Quick Revision Points

- The nucleus contains protons and neutrons.
- Nuclear force is very strong and short ranged.
- Binding energy determines the stability of the nucleus.
- Radioactivity occurs in unstable nuclei.
- Half life is represented by $T_{1/2}$.
- Nuclear fission splits heavy nuclei.
- Nuclear fusion combines light nuclei.

Important Formulae

$$\Delta m = [Zm_p + (A - Z)m_n] - M$$

$$E = \Delta mc^2$$

$$T_{1/2} = 0.693 / \lambda$$